

Climate Resilience: Comparing the Evolution of Integrated Climate Policies

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1 Introduction

The Paris Agreement, an international treaty reached at COP 21 on December 12th 2015 is a global effort to bring all countries together under a common cause. The primary objective is to strengthen the global response to climate change by keeping the rise in global temperatures “well below 2 degrees Celsius” above pre-industrial levels, while actively pursuing efforts to limit that increase to 1.5 degrees.

The agreement requires all Parties to submit “nationally determined contributions” (NDCs), which are individual country plans that must be updated and strengthened every five years.

This project is aimed at exploring the evolution of climate policies before, during and after the signing of the Paris agreement which entered into force on the 4th of November 2016.

The Paris agreement addresses several crucial areas in the climate policy landscape, including adaptation and mitigation efforts. Mitigation policies are those that aim to reduce or prevent the emission of greenhouse gases, while adaptation policies focus on adjusting to the current and expected impacts of climate change. The need for adaptation is becoming increasingly urgent as the effects of climate change become more pronounced and it is essential to understand how countries are integrating adaptation into their policy portfolios alongside mitigation efforts.

Adaptation (Art. 7) – The Paris Agreement establishes a global goal on adaptation – of enhancing adaptive capacity, strengthening resilience and reducing vulnerability to climate change in the context of the temperature goal of the Agreement. It aims to significantly strengthen national adaptation efforts, including through support and international cooperation. It recognizes that adaptation is a global challenge faced by all. All Parties should engage in adaptation, including by formulating and implementing National Adaptation Plans, and should submit and periodically update an adaptation communication describing their priorities, needs, plans and actions. The adaptation efforts of developing countries should be recognized

The goal of this project is to analyze the evolution of climate policies in a selection of countries, with a particular focus on the integration of adaptation measures alongside mitigation efforts. By examining the cumulative stock of climate policies over time, we aim to identify trends and patterns in policy adoption, and to assess how the Paris Agreement has influenced national policy trajectories.

2 Research Question

3 Hypothesis

4 Data & Methodology

The project integrates data from the Climate Policy Database (CPDB), OECD CAPMF, ND-GAIN, and UNFCCC NDCs. The workflow is orchestrated by a master script (`main.R`) to ensure end-to-end reproducibility.

about the CAPMF, there is a hierarchical structure for categorizing policies. The levels are as follows:

Level		Scale		Example Code		What it represents
Level 1		Building Block		LEV1_SEC		Total Sectoral Policies (Aggregated)
Level 2		Module		LEV2_SEC_E_MBI		Market-based instruments in the Energy sector
Level 3		Instrument		LEV3_CARBONTAX_E		The actual Carbon Tax

policy in Energy Level 4 Variable LEV4_CARBONTAX_E_PR The specific price/rate of that tax

5 Results

6 Discussion

7 Limitations

8 Conclusion

9 LLM Declaration